

Origins: thermodynamics

- ★ Second Law of Thermodynamics: Entropy is unlikely to decrease over time
 - systems tend toward states of high entropy

Main idea: the micro-physical explanation

- micro-state = observable state of a system
- micro-state = microscopic, exact particles that comprise of a system

There are many possible micro-states that result in a given macro-state.

It is very likely that any micro-state tends towards a higher entropy macro-state.

→ macro-state never decreases entropy because vast majority of micro states tend towards that

Implication: Universe tending towards state of equilibrium in the long run

- possible that universe will last infinitely long
- for most of the universe's existence, it will therefore be in equilibrium
- Small chance • micro-states might cause momentary deviations: Boltzmann Brains
 - ↗ highly unlikely, but probable if infinite universe

creation of momentary, isolated brains with exact memories + thoughts of a human
The brain's experiences are entirely false.

Conclusion: it is overwhelmingly likely that our thoughts are false and therefore cannot be trusted. (it is unlikely that we are actually in our perceived environments)